XRPL Validator — Daily Engineering Report · 2026-10-10

Period: 2026-10-10 00:00 to 24:00 US Eastern (2026-10-10 04:00 to 2026-10-11 04:00 UTC); ledgers #107,553,312 to #107,575,701. Every count from the ledger and the validator's logs stops at #107,575,701, closed 03:59:51 UTC. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records; Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores. Not available: none.

Status

Normal — coverage outside planned maintenance 100.000% (objective 99.9%), every validation agreed with the network, no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,379 of the 22,390 ledgers the network closed (99.951%); every validation matched the network's validated ledger. Unsigned: 11 during planned restarts.
- **Native engine:** 2 engine receipts recorded (2 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 1 deployment (cycle 168), 4 commits merged into main, 2 engine findings fixed (F418–F419).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-10-23 09:34 UTC.

Service levels

Measure	Value
Ledgers closed by the network	22,390
Validations issued	22,379
Coverage (validations / ledgers closed)	99.951%
Validations matching the network's validated ledger hash	22,379 of 22,379
Unsigned during planned restarts	11
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	1
Basis: the hash the first peer announced as accepted (before cycle 183)	22,379
External agreement, data.xrpl.org (UTC day 2026-10-10)	0.99934 (13 missed of 19,589)

Restarts:

- 04:44:13 UTC: deployment of cycle 168; validations resumed at #107,554,007 (8 ledgers unsigned between the last validation before the restart and the first after it)

Other planned maintenance:

- **107,554,031–#107,554,033 (3): Within cycle 168's start-up: the deployment completed at 04:46:01 UTC and the first health sample, at 04:48:31 UTC, still showed the validator loading its state.**

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Earlier FFI shadow log (libxrpl applied against the validator's mirror; superseded by Step 1): 59 transaction-level differences recorded this day, across 27 ledgers. These were not triaged into receipts.

Ledgers with a receipt in the period (#107,553,312–#107,575,701), by step:

Step	Ledgers with a receipt
Step 1 · native engine	2
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	2

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	22:17 UTC	#107,570,369	Step 1 · native engine, Writer	1 object(s) differed in content	fixed: F418 (a delegation re-filed by name or number), live since 10-11 (cycle 169)
R-02	00:03 UTC	#107,572,028	Step 1 · native engine, Writer	4 object(s) differed in content	fixed: F419 (AMMCreate's trust line order), live since 10-11 (cycle 169)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-10-01, INC-01 is INC-2026-10-10-01.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
168	`863651f`	04:45:02 UTC	1,975 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	419 matched, 0 mismatched	#43 from 05:12 UTC: receipts=2 (2026-10-11 05:12 UTC)

Cycle 168 delivered: `59a0ddc` fix(probe): differential_probe keeps a tecEXPIRED's erasures, as the node does (devnet #5533675 7E7F5D06640B); `66081c8` docs: the plan to a fully independent validator, updated for Stage 4 Phase B; `7cf3aff` docs: the independence plan, in depth; `74a981e` docs: the independence task breakdown and the recommended order; `5710fd4` fix(ledger): fixBatchV1_2 — a Batch inner must be a RawTransaction object (rippled 3.4.1); `3a9c669` fix(flow): rippled 3.4.1's overflow hardening — checked savedIns/savedOuts sums in flow() and BookStep; `863651f` fix(ffi): leg A credits a shared Batch inner to the outer its ParentBatchID names, not the last carrier

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
05:03 UTC	xrpl-validator	`4bb4877`	docs: fixBatchV1_2 is ported (cycle 168), so task 1.5 is done and the known-gap list starts empty
23:00 UTC	xrpl-validator	`452f251`	fix(ledger): F418 — a permission by name and by number is one permission, so re-filing a delegation changes nothing (Mainnet #107570369 EE825DEED447)
23:00 UTC	xrpl-validator	`6bd1612`	fix(probe): an RPC error that does not settle a lookup is a failed fetch, not "absent"
00:19 UTC	xrpl-validator	`2053735`	fix(ledger): F419 — AMMCreate links the AMM, sends the LP tokens, then Amount, then Amount2, as rippled does (Mainnet #107572028 AC9F1F115224)

Findings fixed

- **fix(ledger): F418 — a permission by name and by number is one permission, so re-filing a delegation changes nothing (Mainnet #107570369 EE825DEED447)** (`452f251`, xrpl-validator) A DelegateSet re-filed the two permissions its Delegate object already held (OfferCreate, OfferCancel). Mainnet's metadata has only the sender's AccountRoot: rippled's ApplyStateTable never emits a node whose final content equals its original. The engine wrote the object back with the transaction's PermissionValue names, where the decoded original holds their numbers (8, 9), so the write-back read as material, was …
- **fix(ledger): F419 — AMMCreate links the AMM, sends the LP tokens, then Amount, then Amount2, as rippled does (Mainnet #107572028 AC9F1F115224)** (`2053735`, xrpl-validator) An issuer pooled two of its own tokens (PGold / PCopperLockpick), so all three new trust lines went into its own owner directory, whose last page (0x10) had one slot left. rippled's applyCreate sends the LP tokens before the assets: the LP line took the slot and both asset lines opened page 0x11. Ours funded the pool first and minted last, so the LP line landed on 0x11 and the pages' contents and two HighNodes …

Other engineering changes

- **fix(probe): an RPC error that does not settle a lookup is a failed fetch, not "absent"** (`6bd1612`, xrpl-validator) #107541732 (Payment D8D77D7AB213, PLX -> mXRP -> FARM through two AMM pools): s2 answered the PLX/mXRP pool's ledger_entry {amm} with an error in three runs. rpc() returned the error as data, load_book_pair read it as "no pool", and the replay reported a phantom tecPATH_DRY with a verdict (exit 1). One DX_RPCLOG re-run matched 108/108. RPC_FAILED counted only network failures, so a well-formed JSON error slipped …

Docs and operations

- **docs: fixBatchV1_2 is ported (cycle 168), so task 1.5 is done and the known-gap list starts empty** (`4bb4877`, xrpl-validator)

Independence program

No independence record for UTC day 2026-10-10; records begin 2026-10-20.

Network

Measure	Value
Ledgers closed	22,390 (#107,553,312–#107,575,701)
Mean close interval	3.86 s
Longest close interval	9 s (before #107,575,694)
Closes at least 15 s apart	0
Empty ledgers	0

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Open items

- Amendment fixCleanup3_4_0: activation due ~2026-10-23 09:34 UTC.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-10 05:03	10-10 05:03	xrpl-validator	`4bb4877`	docs: fixBatchV1_2 is ported (cycle 168), so task 1.5 is done and the known-gap list starts empty
10-10 23:00	10-10 23:00	xrpl-validator	`452f251`	fix(ledger): F418 — a permission by name and by number is one permission, so re-filing a delegation changes nothing (Mainnet #107570369 EE825DEED447)
10-10 23:00	10-10 23:00	xrpl-validator	`6bd1612`	fix(probe): an RPC error that does not settle a lookup is a failed fetch, not "absent"
10-11 00:13	10-11 00:19	xrpl-validator	`2053735`	fix(ledger): F419 — AMMCreate links the AMM, sends the LP tokens, then Amount, then Amount2, as rippled does (Mainnet #107572028 AC9F1F115224)

Times in UTC. A commit appears on the day it was committed to main or merged into it.